

ABEL BRIONES RAMIREZ

ENGINEERING & TECHNOLOGY LEADER

Complex Industrial Systems | Automation & Modernization | Optimization & Applied AI | Engineering Leadership
Querétaro, Mexico | Open to relocation: Barcelona/Spain, Germany, Finland | hebelmx@gmail.com | abel.briones@exxerpro.com |
[linkedin.com/in/abel-briones/](https://www.linkedin.com/in/abel-briones/) | <https://hebelmx.github.io/abel-briones/> | Cel 442230211

EXECUTIVE PROFILE

Engineering and technology leader with 25+ years spanning industrial automation, complex machine modernization, motion and control systems, mathematical optimization, software, applied research and emerging AI technologies. Founder of Exxerpro Solutions, an industrial engineering company operated from 2010 to 2026, with MXN 150M+ in cumulative invoiced revenue and a peak team of 21. Particularly effective in technically ambiguous, production-critical situations that require first-principles analysis, evidence-based diagnosis, multidisciplinary coordination and practical execution. Combines hands-on engineering depth with team development, customer leadership, strategic planning and the ability to turn individual expertise into documented, repeatable engineering processes.

CORE VALUE

Complex Problem Solving

Root-cause diagnosis, first-principles decomposition, hypothesis-driven testing

Engineering Leadership

Architecture, technical authority, mentoring, commissioning and crisis decisions

Industrial Systems

Automation, motion, safety, networks, modernization and production systems

Applied AI & Optimization

Modeling, simulation, evolutionary optimization, industrial AI and digital engineering

Business & Customer

P&L responsibility, technical sales, key accounts, scope/constraint negotiation

Research & Innovation

Peer-reviewed research, technology transfer, productization and R&D

Process & Quality

ISO 9001 environment, structured diagnostics, verification, validation and documentation

Cross-Domain Thinking

Connects physical systems, software, data and mathematical models across disciplines

PROFESSIONAL EXPERIENCE

Exxerpro Solutions S.A. de C.V. | Querétaro, México

2010 - 2026

Founder | Engineering & Technology Leader | Strategic Planning Director

- Built and led an industrial engineering and technology company serving automotive, manufacturing, food and personal-care sectors; generated MXN 150M+ in cumulative invoiced revenue.
- Scaled the organization to a peak team of 21; developed engineers and supervisors through mentoring, learning-by-doing and documented methods while retaining technical authority for difficult decisions and escalations.
- Led or oversaw hundreds of projects and specialized services spanning control modernization, motion, networks, safety, vision, traceability, software and production optimization.
- Served as technical authority on production-critical modernization programs from architecture through commissioning and production support, coordinating multidisciplinary teams under tight downtime constraints.
- C-Level & Technical Relationships: Developed strategic partnerships with global organizations including Valeo (Mexico & North America), Rockwell Automation, TREMEC, Dana, Mecalux, 3M, Donaldson, Procter & Gamble, Mabe, Kimberly-Clark, Bachoco, Maxion Wheels, Kerry Ingredients, Unilever, Halyard, DEACERO, Prettl, Xenon Automation, Saint-Gobain, ABB, Aernnova, VRK Automotive, Conduemex, Nacobre, Iritech, Vibracoustic, Maccaferri, and Flex-N-Gate..
- Codified a repeatable engineering methodology for problem definition, evidence gathering, competing hypotheses, risk-controlled experimentation, verification and validation.
- Directed post-pandemic strategic evolution toward specialization, proprietary solutions, digitalization, sustainability, software and applied AI; formalized and communicated the 2025-2035 strategy.

Innovación Integral de Sistemas S.A. de C.V. | Querétaro, México

2002 - 2009

Co-Founder | Board Member | Project Director

- Directed industrial automation and integration projects for manufacturing customers, combining field engineering, project leadership, customer coordination and technical problem solving.
- Connected industrial practice with academic research, applying optimization and computational methods to manufacturing and process-engineering problems.

GUMA Automatización / CFE | Querétaro, México

1998 - 2002

Project Manager | Control Systems Engineer

- Delivered industrial control and visualization projects, coordinating plant personnel and engineering teams while developing hands-on expertise in PLCs, instrumentation, SCADA/HMI and industrial communications.

SELECTED ENGINEERING EVIDENCE

- Industrial modernization:** 150+ control systems across a reconstructed career inventory, including legacy PLC/HMI, motion, safety and VFD applications.
- Production troubleshooting:** 200+ machines and lines involving downtime, recurring failures, cycle-time/takt improvement, emergency recovery and unfamiliar technologies.
- Complex motion systems:** led modernization and diagnosis of multi-axis machinery, including systems with multiple processors and dozens of axes; used kinematic modeling and empirical calibration to resolve synchronization and scaling problems under production constraints.
- Industrial communications:** 40+ network projects across PROFIBUS, DeviceNet, ControlNet, CANbus, PROFINET, EtherNet/IP and Modbus TCP; used packet-level evidence for intermittent faults.
- Digital production systems:** delivered multiple traceability deployments integrating PLC data, barcode/RFID identification and process-variable capture; also developed custom industrial software and monitoring solutions.

RESEARCH, OPTIMIZATION & APPLIED AI

Applied researcher with work in mathematical modeling, simulation, process optimization, evolutionary computation and industrial applications. Research has included constrained multi-objective genetic algorithms, thermally coupled/distillation systems, production-line optimization and surrogate neural-network models used to accelerate computationally expensive simulations.

- Co-authored peer-reviewed research published in Computers & Chemical Engineering and related engineering venues; presented research at international conferences and contributed book chapters.

- Developed and applied multi-objective evolutionary optimization linked with rigorous process simulation; used neural-network surrogate modeling where direct simulation was too computationally expensive.
- Continued the same modeling mindset in industrial control work, using simplified physical/kinematic models, simulation and feed-forward concepts to solve difficult machine and control problems.
- Current direction: industrial AI, LLM/RAG/agent systems and digital engineering, emphasizing explainable, evidence-based assistance for engineers.

REPRESENTATIVE PROBLEM-SOLVING CASES

- Production-critical controls modernization: specified architecture and led technical execution for a major legacy controls and motion migration with no practical fallback once the old system was retired; supervised engineers, resolved motion issues during commissioning and supported first production.
- High-speed machine synchronization: entered a failed motion migration as a diagnostic resource, constructed an empirical mechanical model, identified configuration/scaling errors hidden by compensating patches and restored intended synchronized behavior.
- Intermittent robot/network failure: challenged an incorrect attribution to an unrelated HMI change, instrumented the system with cameras and network capture, isolated connection starvation caused by network configuration and eliminated the recurring failure.
- Remote control-system tuning: reconstructed a control-system model from field information, cross-validated simulation results, detected a unit mismatch, recalculated the model and supplied tuning parameters that worked in the field.

LEADERSHIP & MANAGEMENT

- Leadership style combines directive decision-making during time-critical events with a Socratic coaching approach when time permits; emphasizes process, evidence and development of independent technical judgment.
- Managed full business responsibility across engineering delivery, staffing, customer relationships, technical sales, supplier decisions, investment choices and strategic planning.
- Known for technology-independent recommendations and willingness to reject commercially attractive work when the technical basis, scope or customer value is not defensible.
- ISO 9001:2015-certified company environment; formalized documentation, prioritization, engineering diagnostics, risk assessment, verification, validation and lessons-learned practices.

EDUCATION

Instituto Tecnológico de Querétaro (ITQ) · anabin H+ 2000 - 2004
M.S. Industrial Engineering (Manufacturing)
 • Graduate work focused on optimization of serial production lines using genetic algorithms; 1st Place, National Graduate Thesis Competition (2004).
Instituto Tecnológico de Querétaro (ITQ) · anabin H+ 1995 - 1998
B.S. Electronic Engineering - Instrumentation & Control

CREDENTIALS & TECHNICAL BREADTH

Credentials: SEP-CONOCER Certified Instructor & Course Designer (EC0217.01, EC0301).

Industrial systems: Siemens and Rockwell control ecosystems; PLC/HMI/SCADA; motion/servo systems; industrial networks; safety; machine vision; traceability; VFDs.

Machine Safety & PackML: Designed and implemented safety PLC systems and collaborated on machine risk assessments applying **ISO 12100 and ISO 13849 principles**, including safety functions targeting performance levels up to **PL e**. Developed multiple machine-control applications based on the **PackML** standard.

Software & computation: C#/.NET, Python, SQL Server, mathematical modeling, simulation, optimization algorithms, neural networks, industrial AI, LLM/RAG/agent systems.

Languages: Spanish (native); English (professional).